

Current-generation MLFF campaign CLI specification

Version: mdstats 0.20.242a0 current contract Status: implemented for the P1-P5 functional campaign lifecycle

Purpose and authority

The campaign CLI projects the current source, statistical, target-size, post-selection, and production owners into one restartable user workflow. It does not replace those owners or create a second scientific state machine. Architecture defines ownership and data flow; this specification defines the public command, configuration, persistence, currentness, and failure contract.

The source-checkout entry point is:

```
python tools/mdstats-mlff-campaign.py --config campaign.toml <command>
```

Public command surface

The parser exposes exactly these commands:

```
init
doctor
prepare
select-target-size
cross-validate
train-production
status
advance
guide
storage
```

The scientific lifecycle is:

```
init -> doctor -> prepare -> select-target-size -> cross-validate -> train-production
```

storage is orthogonal artifact management. status and advance derive their projection from the same current owners. The current campaign has no separate pre-screen gate or downstream physical-test command; downstream qualification is a later product boundary and is not dispatched by P6.

No command may silently skip a failed, stale, waiting, or incompatible current authority. An automatic target-size diagnostic that completes without a recommendation is reported as a successful diagnostic result: it leaves the current provisional design unchanged and does not prohibit an explicit target-size choice.

User-visible layout

The operator supplies campaign.toml. The workspace contains:

```
<workspace>/campaign-manifest.json
<workspace>/mdstats/campaign.sqlite3
<workspace>/mdstats/                # current records and reconstructible caches
<workspace>/data/                   # current MACE materializations
<workspace>/runs/                   # authorized checkpoints and logs
<workspace>/models/                 # current production publication
<workspace>/results/                # bounded summaries and cleanup reports
```

SQLite rows and content-addressed files are authoritative only through their own current owners. File existence, a caller-held object, or a cache path is not evidence of currentness.

Configuration contract

campaign.toml owns requested policy and source locations. Runtime observations and durable scientific outcomes are persisted separately. Paths are resolved relative to the configuration file, and a non-positive execution timeout means no campaign wall-clock timeout.

The generated current configuration exposes:

```
[target_data.size_convergence]
target_size_power_min = 7
target_size_power_max = 14
evaluation_size_powers = [8, 9, 10]
fidelity_epochs = [1, 3, 10]
```

```
[post_selection.cv]
fold_count = 2
partition_seed = 7
seeds = [11]
max_num_epochs = 2
acceptance_maximum = 0.5
```

```
[post_selection.production]
seeds = [5]
```

The configured power ceiling is not a fixed scientific constant. Candidates are additionally bounded by the available `P_train` population and the current policy. Optimizer seeds are authored by the sole enabled training method; there is no separate target-size seed namespace. Pre-target fold-count and partition-seed fields are not current target-size/CV authoring controls.

The learned model precision is `single` (FP32) or `double` (FP64). Critical mdstats reductions, geometry/statistical arithmetic, and persistent bookkeeping remain FP64. The acceleration backend, MACE interface, and runtime identity are bound by the doctor/current protocol record.

Policy configuration is fail-closed at the existing semantic owners. Finite real fields reject booleans and strings, integer fields require actual integers, boolean fields require actual booleans, and collection elements are validated before canonical ordering or float conversion. This includes the target-size normalization reference, global objective, and configuration-weight policies. Current-schema readers reject malformed values before identity or execution; only an explicitly supported historical reader may apply a historical representation.

Command behavior

init

`init` writes an annotated configuration with current target-size and post-selection sections. It must not generate retired lifecycle sections or imply an unconfigured target-size ceiling. Existing configuration files are not over-written without the explicit force behavior defined by the parser.

doctor

`doctor` checks source paths, manifest inputs, foundation/replay identity, package/runtime imports, precision wrappers, requested backend, and available resources. A successful result records the exact runtime realization used by later owners. Unsupported hardware or unavailable long-production resources remain an explicit unavailable/deferred condition, never a fabricated pass.

prepare

`prepare` authenticates the manifest and constructs/reuses the current neutral substrate:

```
source/frame/label authority
-> protected statistical relations
-> one P_train/M3 split and pi_train/pi_eval
-> one common target-size preparation
```

It selects no target size, trains no candidate, ranks no checkpoint, and publishes no final model. It is restartable and idempotent when all current inputs and identities match, including when the current generation already carries a complete automatic diagnostic: an unchanged prepare is a successful no-op that leaves the canonical generation, its diagnostic evidence, and its derived result view exactly as they were.

prepare is the sole command permitted to interpret live inputs, so it also owns detecting that they changed. Before reusing the stored lower-level catalog it compares the approved manifest digest that catalog was built from and each source's own byte-identity and control signatures against the files on disk. Materially changed valid inputs are routed through the ordinary catalog reconstruction path and committed as a fresh canonical generation; malformed or unapproved changes still fail under their existing owners. No operator flag is required for this. `--approve-manifest` records the reviewed digest; `--refresh-inferences` refreshes proposed source metadata before approval; `--rebuild-catalog` remains an explicit unconditional reconstruction request, not the mechanism by which ordinary input changes are noticed.

Durable publication is create-or-verify. A prepared component, generation manifest, or normalized frame entry whose content identity already exists on disk is authenticated against that identity – for a frame entry, including every array member's hash, shape, and dtype – before this generation may reuse it. Conflicting or corrupt content fails before adoption and is never overwritten, because another adopted generation may still depend on the bytes that are actually there.

Adoption is compare-and-set against the currentness token captured *before* the expensive construction began, and no campaign-wide writer lock is held across that construction. Two concurrent preparations that produce the same prepared identity converge on one generation; a preparation whose snapshot was built against superseded state fails closed rather than advancing the newer generation.

At the destructive generation boundary, obsolete derived target-size records are detected before semantic decoding, quarantined under a namespace no current loader reads, and rejected rather than migrated. Lower-level source or content caches are reusable only after current-owner revalidation.

select-target-size

```
select-target-size <N>
select-target-size --auto
[--select-horizon-cv <positive integer>] [--select-horizon <positive integer>]
```

This command owns the **provisional** downstream training design and freezes nothing. A bare invocation is invalid and must give actionable usage guidance; `<N>` and `--auto` are mutually exclusive.

`select-target-size <N>` loads the current prepared generation, establishes currentness, resolves the P2 qualified candidate set, requires `N` to be one of those configured qualified candidates, derives `T_N = pi_train[:N]`, validates that membership through the existing P2 training-order owner, resolves a complete provisional horizon snapshot, and CAS-publishes the new proposal. It performs **no** target-size candidate training and no EVAL2 work.

`--select-horizon-cv` sets the provisional CV max epochs and `--select-horizon` the provisional final-production max epochs. Each successful proposal-setting invocation resolves a complete proposal: an omitted flag resolves from `[post_selection.cv].max_num_epochs` (default 30) or `[training].max_num_epochs` (default 30) *at that moment*, and the resolved values are persisted, so later configuration edits do not mutate an existing proposal. An earlier override never becomes a sticky default. The CLI never rewrites `campaign.toml`.

`select-target-size --auto` runs or reuses the optional automatic diagnostic. It runs the configurable ladder and direct evaluation populations through the authenticated continuation

`n1 / M1 -> n2 / M2 -> n3 / M3`

with paired optimizer seeds from the sole enabled method. Each candidate is an exact prefix of the one `pi_train` order. The reducer publishes either a **recommended** `N` with the exact identity of `pi_train[:N]`, or a typed no-recommendation outcome. Replay and later validation evidence cannot affect that computation. A valid recommendation becomes the current provisional `N`.

A completed diagnostic that produces no recommendation must not partially modify any proposal field: the diagnostic evidence and its portable report are committed, any previously valid proposal is unchanged, and the command returns normal success while stating that no recommendation was established.

If a current reusable complete diagnostic already exists for the campaign's automatic-screen scientific/execution identity, `--auto` authenticates and reloads it, performs zero new TRAIN2/EVAL2 work, and says so. Provisional `N`, either horizon flag, either configured horizon, report presentation, status wording, report path, and selection-source provenance never invalidate a current diagnostic.

A diagnostic may run for hours. It captures the selection state it intends to update; if a newer explicit choice or a `cross-validate` freeze intervened, the evidence and report are still published but the stale recommendation is not installed, and the command says so.

Every terminal diagnostic writes one portable, self-contained Markdown report under the campaign `results/` tree at a stable per-generation filename. It is a derived, rebuildable projection of authenticated evidence and is never read as authority.

Candidate learning-rate amplitude and EMA decay are normalized against the configured reference size (`[target_data.size_convergence.optimizer_normalization]`) so a larger candidate does not also receive more optimizer progress; epoch counts, batch size, LR shape, and every other optimizer setting stay fixed across candidates.

The configured ladder ceiling is a practical budget limit. When `Nmax` remains materially superior to every other successful terminal finalist, it is **recommended** and the result carries the non-blocking warning code `nonconverged_at_configured_ceiling`; status, the derived result view, and the portable report state the warning alongside the recommendation. Inside the practical-equivalence band the smaller finalist is still preferred. Genuinely insufficient comparison remains a typed no-recommendation outcome, and no unconfigured intermediate or rescue size is ever synthesized.

cross-validate

This command is the freeze boundary. It requires a current provisional proposal, loads and authenticates the current prepared generation, revalidates the proposed `N` against the current qualified candidate set, derives exact `T_selected = pi_train[:N]`, reproduces its membership digest, and publishes the frozen target selection together with both effective role horizons through the existing currentness/CAS boundaries before any numerical CV work begins. No automatic-diagnostic execution head or reducer is required for this transition. After the freeze, `select-target-size` in either form must not mutate the design.

It then constructs the configured `K >= 2` post-selection folds inside exactly `T_selected`. It binds protected relations, fold/seed identities, target-only checkpoint choice, and the all-required-fold/all-required-seed acceptance predicate. It cannot alter the selected size or membership. Missing, stale, or failed fold evidence blocks final production while leaving the selected authority unchanged.

train-production

This command requires accepted current post-selection method evidence. It starts fresh from the accepted foundation and trains the complete exact `T_selected` under the frozen production horizon and the production policy. A screen or CV checkpoint is never a production parent. Publication rechecks currentness at commit time and cannot promote work from a superseded generation.

status, advance, and guide

`status` projects the current owner states and reports the next safe action. `advance` dispatches only the next current lifecycle owner. `guide` prints the same six-command scientific lifecycle and current configuration/restart semantics. Neither command writes a second scientific authority.

Observation is read-only, coherent, and authenticated. One lifecycle answer reads the target-size revision and every post-selection/qualification pointer row it depends on inside a single campaign-store read transaction, so the ancestry it reports is one that existed: pointer publication mutates campaign metadata without moving the target-size revision, and an answer may never combine a pre-publication view of one stage with a post-publication view of another. Each compact record a pointer names is loaded through its accepted read-only typed store and must reproduce the identity the pointer named before any of its fields – a cross-validation acceptance, a release verdict – influences the report. A missing, unparseable, or misidentified record is reported as blocked. Observation creates no evidence root, opens no provider, and reconstructs no upstream authority.

storage

Storage reports and manages only campaign-owned artifacts. Read-only inventory is available through storage report. Cleanup tiers require their documented dry-run/apply behavior; archive create, integrity check, and restore are reversible and independently content-checked. External inputs, current scientific records, selected checkpoints, restart state, and diagnostic evidence remain protected.

Persistence and restart

CampaignStore shall:

1. use one SQLite database for durable campaign state;
2. serialize canonical payloads with content identities under their owners;
3. record operational stage state without treating it as scientific authority;
4. preserve the latest successful evidence before process exit;
5. retain bounded event history;
6. never use the database location itself as scientific identity.

Current preparation and post-selection owners rederive their input identities on every reopen. A matching completed record is reused only after source, policy, parent, payload, and currentness authentication. A missing, corrupt, stale, or incompatible derived cache is rebuilt by its owner; it is not silently accepted or translated.

Each current write uses a generation-neutral prepare/restart identity. A historical generation marker may be inspected only by the reject-only cutover detector and may not authorize semantic reconstruction. The exact accepted current-generation P5A6 workspace must remain reopenable without a pre-load rewrite; P6 compatibility evidence is separate from fresh P6 restart evidence.

Configuration changes have selective invalidation:

```
target/frame/label/order/ladder/fidelity/common-preparation change
-> new target-size generation and descendants
post-selection CV-only change
-> CV and final-production descendants only
production-only horizon/runtime change
-> final-production descendants only
```

Provenance-only presentation changes do not change scientific arithmetic.

Failure contract

The CLI fails closed for unapproved or changed manifests, missing or incompatible source/foundation/replay inputs, invalid labels, unresolved protected relations, unsupported policy generation, stale current pointers, missing required folds/seeds, corrupt checkpoint/companion state, no admissible checkpoint, target-size lineage mismatch, incompatible persisted state, or a currentness race at publication.

Keyboard interruption returns the parser's documented interruption status and states that authenticated records remain resumable. Errors identify the owning command and the safe corrective action; they do not fall back to a different scientific selector or downstream model.

Acceptance boundary

The current CLI contract is established through the real parser/facade, generated-config parsing, current prepare/selection/CV/production owners, CampaignStore close/reopen, currentness reauthentication, storage cleanup owners, and exact-generation reject-before-reuse tests. The mandatory P5A6-to-P6 qualification additionally authenticates the baseline worktree and import roots before producing state, then opens the unchanged workspace in a separate P6 process.

P6 acceptance is current functional/restart/public-surface closure. It is not GPU, long real-data, M-ladder decision-preservation, deployment, physical, calibration, locked-test, or final-release qualification.